Value putting:

i) No. of variables > no. of eqns.

ii) Formula asked $\begin{cases} v \to v \\ v \to c \end{cases}$

$\theta \to 90°/0°,\ 45°,\ 30°/60°$

$\frac{\theta}{2}, 2\theta \to 60, 30, 15$

iii) Misc.
- Putting all angles equal. (Symmetrical)
- Substituting complementary values.

1. The simplified value of $(secA - cosA)^2 + (cosecA - sinA)^2 - (cotA - tanA)^2$

$(secA - cosA)^2 + (cosecA - sinA)^2 -$

a) 0 b) $\frac{1}{2}$

c) 1 d) 2

2. **The value of $(1 + cot\theta - cosec\theta)(1 + tan\theta + sec\theta)$ is equal to**

a) 1 **b) 2**

c) 0 **d) -1**

(3) If $(sin\alpha + cosec\alpha)^2 + (cos\alpha + sec\alpha)^2 = k + \tan^2\alpha + \cot^2\alpha$, then the value of k is

a) 1 b) 7

c) 3 d) 5

(4) Find the value of $3\left(\sin^4\theta+\cos^4\theta\right)+2\left(\sin^6\theta+\cos^6\theta\right)+12\sin^2\theta\cos^2\theta$

$3\left(\sin^4\theta+\cos^4\theta\right)+2\left(\sin^6\theta+\cos^6\theta\right)+$

a) 3 b) 2

c) 0 d) 5

⑤ The simplified value of $(sec\ x\ sec\ y + tan\ x\ tan\ y)^2 - (sec\ x\ tan\ y + tan\ x\ sec\ y)^2$ $(sec\ x\ sec\ y + tan\ x\ tan\ y)^2 - (sec\ x\ tan\ y + tan\ x\ sec\ y)^2$

a) -1 b) 0

c) $\sec^2 x$ d) 1

6 What is the value of $\frac{[(\sin x+\sin y)(\sin x-\sin y)]}{(\cos x+\cos y)(\cos y-\cos x)}$?

$\frac{}{[\ \sin x+\sin y \quad \sin x-\sin y\ ]}$

a) 0 b) 1

c) -1 d) 2

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7 What is the value of $\sin(B-C)\cos(A-D)+\sin(A-B)\cos(C-D)+\sin(C-A)\cos(B-D)$?

$sin\ B-C\ \cos\ A-D\ +\sin\ A-B\ \cos(C-$

a) $\frac{3}{2}$ b) -3

c) 1 d) 0

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(8) What is the value of $cos(90-B)\,sin(C-A)+sin(90+A)\,cos(B+C)\,-sin(90-C)\,cos(A+B)$?

$cos\ 90-B\ \ sin\ C-A\ \ +sin\ 90+A\ \ cos(B+$

a) 1 b) $sin(A+B-C)$

c) 0 d) $cos(B+C-A)$

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(9) **What is the value of**

$$\frac{[sin(y-z)+sin(y+z)+2sin\,y]}{sin(x-z)+sin(x+z)+2sin\,x}\,?$$

$$[sin\,y-z+sin\,y+z+2sin\,y]$$

a) $cosx\ sin\,y$

b) $\frac{sin\,y}{sin\,x}$

c) $sin\,z$

d) $sinx\ tany$

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10 What is the value of $[(Cos^3 2\theta + 3 Cos 2\theta) \div (Cos^6 \theta - Sin^6\theta)]$?

$[(Cos^3 2\theta + 3 Cos 2\theta) \div$

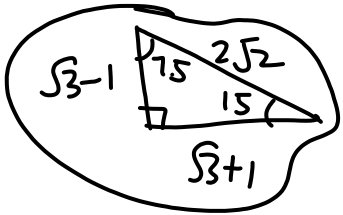
a) 0 b) 1
c) 4 d) 2

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(11) $\frac{\cot\theta}{\cot\theta-\cot 3\theta}+\frac{\tan\theta}{\tan\theta-\tan 3\theta}=?$

a) 1 b) -1

c) 2 d) 0



(12) The value of $\frac{\sin\theta+\cos\theta-1}{\sin\theta-\cos\theta+1} \times \frac{\tan^2\theta(\operatorname{cosec}^2\theta-1)}{\sec\theta-\tan\theta}$ is:

$\frac{\sin\theta+\cos\theta-1}{} \times \frac{\tan\ \theta\ \operatorname{cosec}^2\theta-1}{}$

a) 0

b) -1

c) 1

d) $\frac{1}{2}$

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(13) $\frac{(2sinA)(1+sinA)}{1+sinA+cosA}$ is equal to:

a) $1 + sinA - cosA$

b) $1 - sinAcosA$

c) $1 + cosA - sinA$

d) $1 + sinAcosA$

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(14) What is the value of $\sin(90° + 2A)[4 - \cos^2(90° - 2A)]$?

a) $2(cos^3A - sin^3A)$

b) $2(cos^3A + sin^3A)$

c) $4(cos^6A + sin^6A)$

d) $4(cos^6A - sin^6A)$

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15

HW

If $x = cosec\,\theta - \sin\theta$ and $y = \sec\theta - \cos\theta$, then the relation between x and y is

a) $x + y + 3 = 1$

b) $x^2y^2(x^2 + y^2 + 3) = 1$

c) $x^2(x^2 + y^2 - 5) = 1$

d) $y^2(x^2 + y^2 - 5) = 1$

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16 If $\cot\theta + \tan\theta = p$ and

HW $\sec\theta - \cos\theta = q$ then

$(p^2q)^{\frac{2}{3}} - (pq^2)^{\frac{2}{3}} = ?$

$p^2q^{\frac{2}{3}} - pq^2{}^{\frac{2}{3}} = ?$

a) 0 b) 1

c) 2 d) 3

CDS 2016 I

(17) If $\frac{cos\alpha}{sin\alpha+cos\beta}+\frac{cos\beta}{sin\beta-cos\alpha}=\frac{x}{sin\alpha-cos\beta}+\frac{cos\beta}{sin\beta+cos\alpha}$ then 'x' is equal to:

$$\frac{cos\alpha}{\quad}+\frac{cos\beta}{\quad}=\frac{x}{\quad}+\frac{cos\beta}{\quad}$$

a) $cos\beta$ b) $cos\alpha$

c) $sin\beta$ d) $sin\alpha$

SSC CHSL 2018

(18) What is the value of $\left[(sec\,2\theta + 1)\sqrt{sec^2\theta - 1}\right] \times \frac{1}{2}(cot\ \theta - tan\ \theta)$?

$\left[(sec\,2\theta + 1)\sqrt{sec^2\theta - 1}\right] \times \frac{1}{}$

a) 0 b) 1

c) $cosec\ \theta$ d) $sec\ \theta$

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19 If $sinA + sinB = C$ & $cosA + cosB = D$, then $sin(A + B) = ?$

a) CD

b) $\frac{CD}{C^2+D^2}$

c) $\frac{C^2+D^2}{2CD}$

d) $\frac{2CD}{C^2+D^2}$

20 If $\alpha + \beta = 90°$ then the value of $(1 - \sin^2 \alpha)(1 - \cos^2 \alpha) \times (1 + \cot^2 \beta)(1 + \tan^2 \beta)$ is

a) 1 b) -1

c) 0 d) 2

21 HW

If $\angle A$ and $\angle B$ are complementary to each other, then the value of $\sec^2 A + \sec^2 B - \sec^2 A \sec^2 B$ is

$sec\ \ B - sec\ \ A\, sec\ \ B = ?$

a) 1 b) -1

c) 2 d) 0

22 HW The value of the expression $\left(cos^6\theta + sin^6\theta - 1\right)\left(tan^2\theta + cot^2\theta + 2\right)$ is:

a) 0 b) -1

c) -3 d) 1

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(23) $tan\left(\frac{\pi}{4}+\theta\right)-tan\left(\frac{\pi}{4}-\theta\right)=?$

a) $2\ tan\ 2\theta$ b) $2\ cot\ 2\theta$

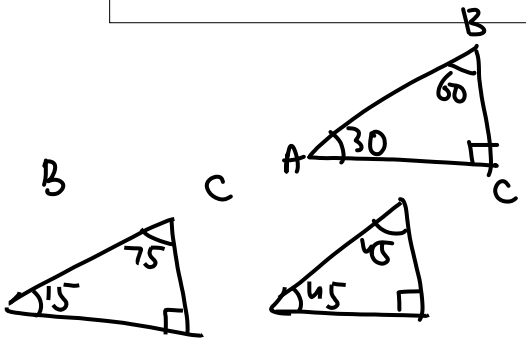
c) $tan\ 2\theta$ d) $cot\ 2\theta$

(24) If A, B, C are angles of a triangle, then $sin^2A + sin^2B + sin^2C - 2cosA\ cosB\ cosC = ?$

$2cosA\ cosB\ cosC = ?$

a) 1 b) 2

c) 3 d) 4



25 If $A + B + C = \frac{3\pi}{2}$, then $cos2A + cos2B + cos2C = ?$

3π

$cos2A + cos2B + cos2C = ?$

a) $1 - 4\,sinA\,sinB\,sinC$

b) $1 - sinA\,sinB\,sinC$

c) $1 - 2sinA\,sinB\,sinC$

d) $1 - 4cosA\,cosB\,cosC$